

Python Quick Reference Notes (Based on Your Questions)

print() vs return

- `print()` displays output on screen.
- `return` sends a value back to the caller.
- Code after `return` inside the same function does not execute.
- Use `print()` to show values; use `return` to reuse values.

Formatting Float Precision

- Use `round(value, 2)` to round to 2 decimal places.
- Use `f'{value:.2f}'` for formatted output.
- Example: `f'{3.14159:.2f}'` gives 3.14.

Lambda Functions

- Anonymous (nameless) functions.
- Syntax: `lambda arguments: expression`
- Can contain only one expression.
- Useful for short operations such as sorting.

***args**

- Accepts any number of positional arguments.
- Arguments are stored as a tuple.
- Example: `def func(*args): ...`

****kwargs**

- Accepts any number of keyword arguments.
- Arguments are stored as a dictionary.
- Example: `def func(**kwargs): ...`

Generators and yield

- `yield` produces values one at a time.
- `yield` pauses the function and remembers its state.
- The generator ends when it reaches the end of the function or encounters `return`.
- Useful for large datasets and infinite sequences.

Your Generator Error

- Problem: return i inside a loop returned an integer immediately.
- for num in even(10) expected an iterable but got an int.
- Fix: replace return with yield.

Scope (LEGB Rule)

- L = Local scope
- E = Enclosing scope
- G = Global scope
- B = Built-in scope

Closures

- An inner function remembering variables from its outer function.
- Works even after the outer function has finished execution.
- Closures use the enclosing scope.